

Week 4 — Risk, Requirements and Prototyping

CSE5IDP Industry Development Project · Presenter: Scott Mann · Online cohort notes

Semester 2, 2026 · Milestone 2 · Trade show day

Worth noting up front: this session ran on trade show day itself (Thursday 6 August, between the 11:00–13:00 and 19:00–21:00 windows), so Scott deliberately cut it short and the content is a bit compressed. The tender document Q&A at the end is arguably the most valuable part.

What was covered

1. Trade show admin

The morning window ran well; the evening is historically busier. Booth editing was re-enabled until 6:30pm that day. Not every team member needs to be present simultaneously.

2. Risk management (continued from W3)

The four-step process was completed: identify, analyse (likelihood times consequence), plan (avoid, then mitigate), monitor. Scott repeated his strong recommendation that one team member take on a dedicated risk manager role. The commercial framing:

$$\text{Risk exposure} = P(\text{occurrence}) \times \text{Impact}$$

Reporting exposure in dollar terms is what gets management attention. In IDP the analogue is wasted project owner hours. He named team availability and productivity as by far the biggest risk in this subject. Tracking can be a Jira plugin or a spreadsheet template — the only real requirement is that the whole team can see it (SharePoint, OneDrive, Teams).

3. ICT project management

Software is intangible IP, so it needs metrics and milestones built deliberately. He warned against reusing a template from a previous similar project and missing what makes the new one unique. Parkinson's law was cited as the reason he favours one-week sprints — fail fast. Common risks named: requirements inflation, student turnover before census date (two online teams have already had significant reductions), poor productivity, specification breakdown.

He also flagged that he had received roughly six emails in two days about intra-team conflict, and that one scrum master was near breaking point over non-responsive teammates.

4. Requirements engineering, IDP-style

Business requirements, then feasibility study, then user requirements, then validate by prototyping. Crucially: **no SRS document is required**. Your Jira board and backlog are the documentation evidence.

The Sommerville mental-health patient management case study was used to show that decomposing a single user story yields five distinct pieces of information — data inputs, outputs, formats, relationships, and likely UI — which then map cleanly onto team specialists (front-end, database, logic layer, with cyber overlaying ISO, IEEE, ASD and OWASP standards across all three). He wants this decomposition happening in every sprint planning meeting.

Five properties of a good requirement: **valid, consistent, complete, realistic, verifiable**. Verifiability

means quantifying things like transaction times or authentication behaviour rather than leaving fuzzy English.

5. Ethnography

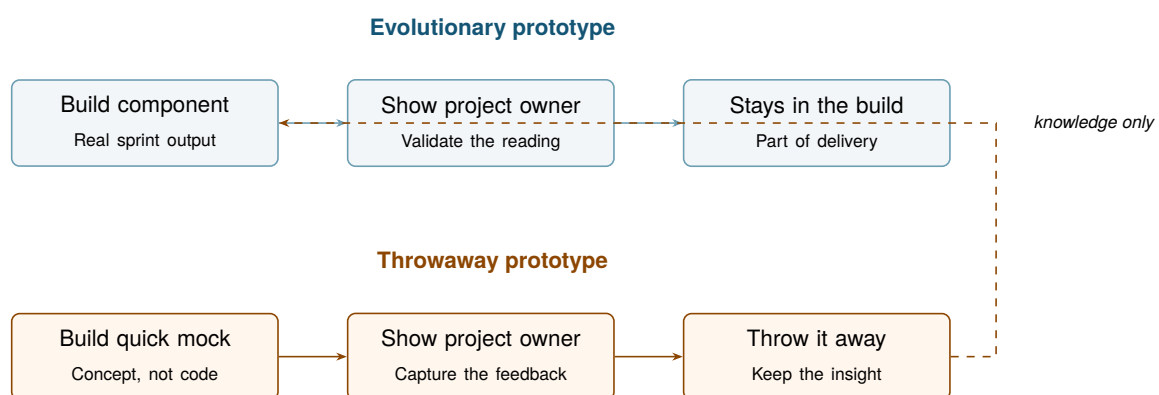
Go to site, observe real users. He called out the Leafy AI teams specifically — get into the labs, see the crops, talk to the scientists. His argument is that AI-assisted development plus genuine domain understanding lets small teams beat consultancy platform lock-in with bespoke solutions.

6. Requirements change management

Triggers: code flaws, regulatory non-compliance, cyber breaches in similar systems, competitor feature releases. The process is identify problem, analyse and specify the change, model the impact across the system, cost it, implement, update the requirement set. His emphasis: never make a snap change to a codebase, because a badly-reasoned fix is often worse than the original bug.

7. Prototyping (the main new concept)

Two forms, and they are not interchangeable.



Only the knowledge crosses from the bottom lane back into the top one — never the artefact itself, because throwaway prototypes are deliberately badly implemented. Scott's stated expectation is that by week 4 most teams have produced one or two **evolutionary** prototypes, and that every sprint ends with something demonstrable.

8. Scrum master presentations

He was explicit that he marks communication, not progress. If the sprint was awful, explain why it was awful. Jira showing "done" is not evidence — he wants to see the actual output, and he framed this as the same assumption your project owner makes: if they cannot see it, it did not happen.

9. Tender document (Q&A)

This is where the real medium-term substance sits, and next week's lecture will expand on it. See the medium-term action items below.

Immediate action items

Trade show submission — verify the date urgently. In W1 Scott stated submissions were due 9 August. He did not restate the date in W4, so confirm against the LMS submission link before anything else. What you submit:

- Only the **Q&A portion** of each recorded interaction, not the full 15–20 minute conversation and not your PowerPoint walkthrough. Trim the video down.
- Recordings from Zoom, Teams, or whatever conferencing tool your booth used.

Padlets on the LMS. Two are live — one on risk scenarios you have encountered, one on the significance of requirements validation. Scott asked directly that people post before next week and said he would work it into the W5 conversation. This is low-effort and he is clearly watching it.

Confirm your scrum master rotation. Only one scrum master per sprint, and only one person from your team presenting on any given lab day. If your team has not nominated for sprint 1, do it now.

Medium-term action items

Tender document — roughly 23 August. He said “17 and a bit days” from 6 August. He hinted at possible flexibility of a few days at most, but warned that extensions compress the back half of the semester. Chapters 1–6 are what is marked:

Section	What he wants
Evaluation criteria and requirements	Row by row against the project requirements — this is where you demonstrate superior understanding versus competing teams
Budget and price	Free-tier components where possible; in-kind human resource cost at the Australian postgraduate graduate rate of \$49.20/hr times hours per sprint. Forecast token, cloud and hardware costs
Constraints, risks and limitations	Your four-step risk workflow in tabular form
Deliverables, hours, timelines	Organise by major feature or user story , not sprint-by-sprint. A Gantt chart is fine
Statement of work	Half to one page — duties and tech stack mapped to user stories
Team members	Skills, experience, qualifications; skill diversity is explicitly rewarded

There is a template on the LMS plus a separate submission link carrying the rubric. He confirmed you may extend and modify the template — he actively encourages it.

Assign a risk manager if your team has not. He offered to sit down with risk managers in labs (Friday face-to-face, Wednesday night with Ali for the online cohort) and said grades are measurably better for teams that do this.

Preserve everything. Research, design work, documentation — it feeds directly into the System Maintenance Document and user manual in the back half of the semester.

Consider use case diagrams. He noted disappointment that few teams used them at the trade show and called them the clearest way to communicate what a system does.

Points he emphasised most heavily

The strongest recurring theme was **do not be a stranger to your project owner**. He offered to personally make phone calls and schedule meetings if you are struggling to get engagement, and framed the alternative as spending a month writing code that gets rejected.

Second was the **budget-versus-quality framing on the tender**: the cheapest team does not win, the team that has thought hardest about the budget does. The same logic applies to the timeline section — feature outcomes, never waterfall steps.

Third, on **AI usage**: he is positive about it as a productivity multiplier for prototyping and generating acceptance criteria, framed as levelling the playing field against consultancy platform lock-in. But this sits alongside his W1 position that it is not a substitute for your own reasoning.

One thing worth flagging: he explicitly noted that team disharmony is on the rise this semester and that census-date departures are already hitting online teams. If your team is carrying any of that, raising it early with Ali is the path he is signposting.

Next week (W5) covers incremental development and a detailed tender document walkthrough.